Annotated explanation of the definition of a full-rank lattice

We are given the statement:

$$\text{A (full-rank) lattice } \Lambda \subset \mathbb{R}^n \text{ is } \Lambda = \{B\mathbf{z} : \mathbf{z} \in \mathbb{Z}^n\}$$

for some basis matrix $B \in \mathbb{R}^{n \times n}$.

This means the following.

1. What is B?

The matrix $B \in \mathbb{R}^{n \times n}$ is an $n \times n$ real matrix whose columns are

$$B = \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \end{bmatrix}.$$

So the columns of B are vectors

$$\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n \in \mathbb{R}^n.$$

Since B is called a *basis matrix*, its columns are linearly independent. Hence they form a basis of $\mathbb{R}^n$.

2. What is $\mathbf{z} \in \mathbb{Z}^n$?

The vector $\mathbf{z}$ is an integer vector:

$$\mathbf{z} = \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{pmatrix}, \qquad z_1, z_2, \ldots, z_n \in \mathbb{Z}.$$

So each coordinate of $\mathbf{z}$ must be an integer.

3. What does $B\mathbf{z}$ mean?

When we multiply B by $\mathbf{z}$, we get

$$B\mathbf{z} = z_1\mathbf{b}_1 + z_2\mathbf{b}_2 + \cdots + z_n\mathbf{b}_n.$$

So $B\mathbf{z}$ is an *integer linear combination* of the columns of B.

4. What is the set $\Lambda = \{B\mathbf{z} : \mathbf{z} \in \mathbb{Z}^n\}$?

This means:

$$\Lambda = \{z_1\mathbf{b}_1 + z_2\mathbf{b}_2 + \cdots + z_n\mathbf{b}_n : \ z_1, \ldots, z_n \in \mathbb{Z}\}.$$

So Λ is the set of all points you can get by taking integer combinations of the basis vectors $\mathbf{b}_1, \ldots, \mathbf{b}_n$.

5. Why is it called a lattice?

Because the resulting set is a discrete, regularly repeated grid of points in $\mathbb{R}^n$.

It is like the usual integer grid $\mathbb{Z}^n$, except it may be stretched, rotated, or skewed by the matrix B.

6. Why "full-rank"?

The phrase *full-rank* means that B has rank n. Equivalently, its columns are linearly independent.

Therefore the lattice is not trapped in a lower-dimensional subspace. It lives in all n dimensions of $\mathbb{R}^n$.

7. Compare with the real span

If we allowed real coefficients instead of integer coefficients, we would get

$$\{B\mathbf{a} : \mathbf{a} \in \mathbb{R}^n\} = \mathbb{R}^n$$

because B is invertible.

But in the lattice definition we only allow integer vectors $\mathbf{z} \in \mathbb{Z}^n$, so we get only a discrete subset:

$$\Lambda = \{B\mathbf{z} : \mathbf{z} \in \mathbb{Z}^n\}.$$

Thus:

- real coefficients give a whole continuous space,
- integer coefficients give a discrete grid of lattice points.

8. Standard example

If

$$B = I_n,$$

the identity matrix, then

$$\Lambda = \{I_n\mathbf{z} : \mathbf{z} \in \mathbb{Z}^n\} = \mathbb{Z}^n.$$

So the standard integer lattice is the simplest example of a full-rank lattice.

9. Example in $\mathbb{R}^2$

Let

$$B = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}.$$

Then its columns are

$$\mathbf{b}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \qquad \mathbf{b}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The lattice is

$$\Lambda = \{B\mathbf{z} : \mathbf{z} \in \mathbb{Z}^2\}.$$

If

$$\mathbf{z} = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}, \qquad z_1, z_2 \in \mathbb{Z},$$

then

$$B\mathbf{z} = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} 2z_1 + z_2 \\ z_2 \end{pmatrix}.$$

Therefore

$$\Lambda = \left\{ \begin{pmatrix} 2z_1 + z_2 \\ z_2 \end{pmatrix} : z_1, z_2 \in \mathbb{Z} \right\}.$$

This is a skewed infinite grid in the plane.

10. Plain English summary

The definition

$$\Lambda = \{B\mathbf{z} : \mathbf{z} \in \mathbb{Z}^n\}$$

means:

> Choose n linearly independent vectors in $\mathbb{R}^n$. Then take all possible whole-number combinations of those vectors. The set of resulting points is the lattice.